INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

REFUSE CONTAINER, HOISTING UNITS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers refuse containers suitable for use with rear truck mounted hoist-haul mechanisms covered in MIL-T-11663, of the make and model as specified (see 6.1).

1.2 Classification. Refuse containers will be of the following types and sizes, as specified (see 6.1): (See figure 1 for illustrations).

Type I - Bottom discharge, rectangular top.

Size 4 - 4 cubic yards (cu yd) net volume.
Size 6 - 6 cu yd net volume.
Size 8 - 8 cu yd net volume.
Size 10 - 10 cu yd net volume.
Size 12 - 12 cu yd net volume.

Type II - Standard bottom discharge.

Size 4 - 4 cu yd net volume.
Size 6 - 6 cu yd net volume.
Size 8 - 8 cu yd net volume.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document improvement Proposal (DD) Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3990

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Type III- Bottom discharge, rounded top.

Size 6 - 6 cu yd net volume.
Size 8 - 8 cu yd net volume.
size 10 - 10 cu yd net volume.

Type IV - Top discharge (tilt).

Size 2-1/2 - 2-1/2 cu yd net volume.
Size 3 - 3 cu yd net volume.
Size 4 - 4 cu yd net volume.
Size 6 - 6 cu yd net volume.
Size 7-1/2 - 7-1/2 cu yd net volume.

Type V - Skip, flat bottom.

Size 4 - 4 cu yd net volume.
Size 6 - 6 cu yd net volume.
Size 8 - 8 cu yd net volume.

1.2.1 Part or identifying number (PIN). A specification based PIN to identify types and sizes of refuse containers covered by this document is included in section 6.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.3 Specifications and standard. The following specifications and standard Form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

MILITARY

MIL-C-l474 - Caster, Industrial.
MIL-T-11663 - Truck, Materials Handling, Container Hoisting: Near Hoist, 4 x 2.
MIL-S-15083 - Steel Casting.

STANDARD

MILITARY

MS 24380 - Casters, Industrial.

(Unless otherwise indicated, copies of federal and military specifications and standard are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.1.2 Other Government document. The following other Government document form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Title 29, CFR, Part 1910.145 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD) adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which is current on the date of the solicitation (see 6.1).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI Z245.3 - Safety Requirements for Stability of Refuse Bins.

(Application For copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Commercial Packing.
ASTM D 3953 - Flat Steel Strap and Connectors.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AMS)

AWS D1.1 - Structural Steel Welding Code.

(Application for copies should be addressed to the American Welding Society, 550 N. W. LeJune Road. P.O. Box 351040. Miami, FL 33135.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.1).

3.2 Standard commercial product. The refuse container shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the refuse container being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.3.1 Steel plate, shapes, and sheet. Unless otherwise specified (see 6.1), steel plate, shapes, and sheet shall be any standard commercial grade OF steel which will satisfy the strength and performance characteristics specified herein.

3.3.2 Cast steel. Steel casting shall conform to MIL-S-15083, grade CW except that castings of alloy steel having greater strength and improved welding characteristics may be utilized.

3.4 Design. The container shall be designed to obtain the specified volume, to withstand all stresses expected to be encountered during normal, heavy usage including those encountered in the test loadings specified herein, and to be completely compatible for use with the hoist-haul mechanisms specified herein (see 1.1 and 6.1). The containers shall be of such design that complete discharge of contents can be performed by the driver using controls located in the cab of the truck. Allowable unit stresses, and the design of all connections and weldments shall be in accordance with accepted engineering practice, and shall incorporate the following factors:

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a. Each lifting pin and its attachment to the container shall be designed to withstand a pull of not less than 7,500 pounds (lb) applied as a shearing action at right angles to the hub. The lifting pins shall be suitably positioned on each end of the container in order to provide proper balance and operation for hoisting and dumping under all normal operating conditions.

b. Each bail bar or bail assembly and its attachments to the container shall be designed to withstand a pull of mot less than 9,000 lb without failure or permanent distortion of the bail assembly or the container. The bail bars shall be designed and positioned to permit ready engagement and safe operation utilizing the automatic dumping hooks on the hoist-haul mechanisms specified.

3.4.1 Limiting dimensions. The width (outside to outside of end walls) or main or lower section for all containers shall be 84 inches ±1/2 inch, and be compatible for use with the hoist-haul mechanisms specified. In addition, the refuse containers shall conform to the requirements and dimensions shown in table I.

TABLE I. Limiting dimensions.

Type and Size	Body shell height +1/2 in. -3 in.	Thickness (inches, not less than)						
		Main or lower section				Upper section		
						Doors		
		Bail wall	Opp wall	Ends	Bottom	Faces	Top	End
Type I								
Size 4	70	1/4	3/16	3/16	1/4	0.1345	0.0598	0.1046
Size 6	72	1/4	3/16	3/16	1/4	0.1345	0.0598	0.1046
Size 8	72	1/4	3/16	3/16	1/4	0.1345	0.0598	0.1046
Size 10	80	1/4	3/16	3/16	1/4	0.1345	0.0598	0.1046
Size 12	84	1/4	3/16	3/16	1/4	0.1345	0.0598	0.1046
Type II								
Size 4	42	1/4	1/4	1/4	1/4		0.0598	
Size 6	53	5/16	1/4	1/4	5/16		0.0598	
Size 8	64	5/16	1/4	1/4	4/16		0.0598	
Type III								
Size 6	71	1/4	3/16	3/16	1/4	0.1046		0.1046
Size 8	77	1/4	3/16	3/16	1/4	0.1046		0.1046
Size 10	87	1/4	3/16	1/4	1/4	0.1046		0.1046

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TABLE I. Limiting dimensions. (continued)

Type and Size	Body shell height +1/2 in. -3 in.	Thickness (inches, not less than)						
		Main or lower section				Upper section		
						Doors		
		Bail wall	Opp wall	Ends	Bottom	Faces	Top	End
Type IV								
Size 2-1/2	36	1/4	1/4	1/4	1/4		0.0598	
Size 3	36	1/4	1/4	1/4	1/4		0.0598	
Size 4	45	1/4	1/4	1/4	1/4		0.0598	
Size 6	58	5/16	1/4	1/4	5/16			
Size 7-1/2	71	5/16	1/4	1/4	5/16			
Type V								
Size 4	45	5/16		5/16	5/16			
Size 6	56	5/16		5/16	5/16			
Size 8	67	5/16		5/16	5/16			

3.4.2 Openings. Openings for loading and discharge shall have an unobstructed area not less than that shown in table II. Bottom discharge area shall be measured with bottom opened full for clumping.

TABLE II. Size of container openings.

Opening	Size not less than
Bottom discharge and sump (types I, II, and III), and open top (types II and IV).	85 percent of the largest horizontal cross-section area of container.
For top doors (types I, II, and IV).	55 percent of horizontal cross-section area of container at top.
For end doors, upper section (types I and III).	24 x 24 inches for size 4. 29 x 29 inches for sizes 6 and 8. 35 x 35 inches for sizes 10 and 12.

3.5 Construction.

3.5.1 Type I. Type I container shall be totally enclosed with rectangular upper and lower sections and with vertical walls. Container shall have two spring-counterbalanced top doors and one door in each offset end of the upper section. Container shall have bottom discharge door. When specified (see 6.1), the bottom discharge section shall be furnished with sump.

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3.5.2 Type II. Type II container shall be bottom discharge, rectangular in shape, with vertical walls and with open top; except that when specified (see 6.1), the unit shall be furnished with two spring-counterbalanced top doors. Container shall have bottom discharge door. When specified (see 6.1), the bottom discharge section shall be furnished with sump.

3.5.3 Type III. Type III container shall be totally enclosed with a rectangular lower sections with vertical walls, and rounded top upper section. Container shall have one door in each offset end wall of the upper section. Container shall have bottom discharge door. When specified (see 6.1), the bottom discharge section shall be furnished with sump.

3.5.4 Type IV. Type IV container shall have vertical end walls and bail wall, and sloping wall on dump side so that the top area is equal to approximately 180 percent of the bottom area. Container shall have open top or when specified (see 6.1), shall have one or two spring-counterbalanced top doors, as specified (see 6.1). The container shall be so proportioned or equipped to prevent overturning, when loaded level, and resting on level ground.

3.5.5 Type V. Type V container shall have vertical end and bail walls, flat bottom, open top, and open Front opposite the bail wall. The edges of the end walls at the front openings shall be sloped, and the area of the bottom shall be equal to approximately 175 percent of the total area of the top.

3.6 Components.

3.6.1 Container shells. The container shell shall be as specified herein, with limiting dimensions and material thickness as shown in table I for applicable types and sizes. The main or lower sections and bottoms shall be of steel plate, the upper sections and doors shall be of sheet steel.

3.6.2 Reinforcements. The refuse container shall be fully reinforced to withstand the stresses from heavy loading and rough usage common to such containers. The ends of angles, channels, or pipes shall have streamlined or capped closures to prevent entry of vermin or foreign objects. Reinforcement shall include the following:

a. Steel angle or channel stiffeners welded vertically on inner surfaces of bail walls to coincide with bearing members of hoist-haul mechanisms.

b. Steel channel stiffeners shall be welded to outside bottom of containers, bottom discharge sections, and bottom discharge sump sections. Channels shall be in sufficient quantity to prevent distortion of the bottom when container is fully loaded and resting on high center or uneven ground, and shall be suitably spaced for proper bearing and balance of the container on the bed of the hoist-haul mechanism.

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c. The perimeters of bottom and top discharge openings shall be reinforced with steel bars, plates, or angles so as to prevent bending or distortion of edges around openings during nomally rough usage common to these containers.

d. Edges of top and end door openings shall be suitably reinforced on the inside to provide rigidity and act as doorstops.

e. The larger doors shall be deformed or braced as required to provide rigidity and prevent distortion under normally rough usage.

3.6.3 Bottom discharge sections. Bottom discharge sections shall be hinged to the bottom of the bail wall and shall be of steel plate. The bottom section shall be reinforced with gussets or plates at hinge points to prevent distortion during use. The end walls of the bottom discharge section or sump section may be of overlapping design, or the end walls may be cut in an inverted "V" pattern from the end wall plate of the main container shell, as appropriate to the manufacturer's standard practice. When required to assure alignment of discharge section with the container shell, steel plate guides shall be welded along end wall contact edges.

3.6.4 Bottom discharge sump section. The bottom discharge section shall be constructed in the form of a liquid-tight sump having a reservoir depth of not less than 5 inches when container is setting in a level position. The capacity of sump shall be as appropriate for the container size based on the 5-inch reservoir depth. The sump and seals shall be so constructed that leakage will not occur when the container, with sump reservoir filled with water to the 5-inch depth, is resting on the ground or in the tilted carrying position on the hoisting unit.

3.6.5 Doors and latches. Steel doors shall fit openings shown in table II. Doors shall have spring-loaded, heavy-duty, steel latches with handles operable from both outside and inside the container. Thickness of metal in doors shall be not less than that shown in table I. Latch bars shall be spring-loaded.

3.6.5.1 Sliding end doors. When specified (see 6.1), type I containers shall have sliding door assembles in the upper portion of each end panel. Doors and tracks shall be designed to operate on the outside. Thickness of sheet steel for end doors shall be not less than 0.1046 inch (U.S. revised standard gage No. 12) nominal thickness. Bottom edge of end doors shall not be over 48 inches above the bottom of the container.

3.6.6 Hinges.

3.6.6.1 Bottom. Hinges for bottom discharge sections and sumps shall be heavy-duty steel forged or plate hinges having steel hinge pins not less than 1 inch diameter. Not less than three hinges shall be provided for each bottom discharge section, and additional hinges shall be provided if required for strength and rigidity.

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3.6.6.2 Top. Hardware for top doors shall consist of heavy-duty steel hinge pins, hinge straps, and counterbalancing coil springs. The spring arrangements and torsion shall be such that the door, when unlatched will rise only approximately 4 inches due to the spring action, and will not pop up to further height and cause possible injury to personnel. The door shall require manual assistance to raise to the half-open position and then shall be spring controlled to full open position or higher when tilted in the dumping position for top discharge containers.

3.6.6.3 End. Hinges for end doors shall be heavy-duty steel plate hinges with fixed hinge pins.

3.5.7 Boom rests. Containers shall have one boom rest on each end wall, located adjacent to the bail wall, to support the lifting beams of the hoist-haul mechanism and secure the container while hauling. Boom rests shall be fabricated of 1/4-inch steel plate or steel bar with reinforcing gussets, and when welded to the container shall project approximately 5-1/2 inches out from the container wall. Types IV and V containers shall have one combination boom rest-chain guide on each end wall adjacent to the boom rest, except that the outer extremity shall have a lip projecting upward approximately 3/4 inch above the apex of the boom rest, to guide the chain during top-discharge operations.

3.6.8 Lifting pins. All containers shall have lifting pins designed in accordance 3.3(a) with dimensions which will permit use with the counterbalanced lifting links supplied the hoist-haul mechanisms specified. Lifting pins shall be steel castings or steel plate and alloy steel shaft suitable for intended use. The hub shafting shall be 2 inches in diameter and approximately 2-1/2 inches long. A horizontal wing shall be welded to the outer end of the hub, and shall be of such dimensions as to pass through the slotted hole in the lifting link of the hoist-haul mechanisms. The wing shall prevent disengagement of the lifting link in all lifting and dumping positions. The lifting pins shall project approximately 4-1/4 inches out from the end wall of the container. The mounting base of the lifting pin shall be proportioned to distribute the design load over a sufficient area of the end wall to reduce local stresses and distortion. Two lifting pins shall be supplied on each container.

3.6.9 Bail bars and assemblies. All containers shall have solid steel bails designed and positioned in accordance with 3.4(b). Unless otherwise specified (see 6.1), types IV and V containers shall have bails recessed near the bottom of the bail walls for use with double hook hoist-haul mechanisms, and the bail for type IV containers shall be pivoted at the bottom with torsion spring to return the bail to the normal position. When specified (see 6.1), types IV and V containers shall have recessed bail bar assemblies suitable for use with old type single hook hoist-haul mechanisms.

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3.6.10 Casters. When specified (see 6.1), casters shall he furnished. When casters are specified, two rigid and two swivel all steel casters conforming to MS 24380 and MIL-C-7474 shall be furnished for each container. The casters shall be constructed to withstand a static load of not less than welded to the bottom of the container so that the casters will clear top of fenders on hoist-haul mechanism when the container is in the haul position. The spacers shall be positioned to center on the supporting members of the hoist-haul mechanism. The casters shall be suitably located to clear the container rest members of the hoist-haul assembly when the container is in the haul position.

3.7 Performance. In addition to other design and operational requirements specified herein, all containers shall be capable of withstanding tests specified in 4.4 without overturning, failure, breakage, or permanent distortion of any part. The test loads for each size container shall be as specified in table III.

TABLE III. Test loads.

Size, as applicable	Weight of test load (lb) (minimum)
2-1/2	4,000
3	5,000
4	7,000
6	9,000
7-1/2 or 8	9,000
10	9,000
12	9,000

3.8 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.1). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer’s current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.9 Stability. The containers shall be stable and be in accordance with ANSI 1245.3.

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3.10 Signs.

3.10.1 Caution signs. Caution signs shall be in accordance with ANSI Z245.3.

3.10.2 Danger Signs. danger signs shall be permanently affixed to the lower center of the outside face of the wall opposite the bail wall and each end wall of the container. The exterior size of danger signs shall be not less than 7 inches high and10 inches long. The sign shall read as follows:

DANGER

STAY CLEAR AT ALL TIMES

WHEN CONTAINER IS OFF GROUND

The configuration, wording, dimensions, and color of the sign shall conform to applicable portion of OSHA Title 29 CFR 1910.145.

3.11 Workmanship.

3.11.1 Steel fabrication. Steel used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Re-entrant cuts shall be made in the best possible manner. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. Burned surfaces of flame cut material shall be ground or machined sufficiently to remove ash and checks. All joints, openings, and doors shall be square and neat appearing. The finished container shall be free from sharp edges, fins, burrs, and sharp projections.

3.11.2 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be continuous for all major joints, and shall have the strength required to satisfy the design, use, and loading conditions. Joints between plates shall be welded on both sides when necessary to withstand the load stresses. Bail bars shall overlap container wall not less than 4 inches on each end and shall be welded solid to the container wall. Intermittent welds may be used for attaching guides and reinforcement members provided such weldments will satisfy the design. Sequency of welding shall be such as to minimize the residual stresses in the steel. Members to be welded shall be positioned and held by jigs or fixtures when necessary to insure accurate alignment. Welding procedures shall be in accordance with the AWS D1.1 and using AM recommendations as to the electrodes, types of welds, and welding processes

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utilized. Before assigning any welder to manual welding work, the manufacturer shall certify to the contracting officer that the welder has passed the qualification tests as prescribed by the AWS for the type of welding operations to be performed.

3.11.3 Castings and forgings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which might reduce the casting ability to perform its intended function. Forgings shall be uniform in quality and condition, and shall be free from tears, cracks, seams, laps, internal ruptures, imbedded scale, segregations, or other defects which may detrimentally affect the suitability for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all rerquirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it conmit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

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4.2.1 First article inspection. The first article inspection shall be performed on one refuse container when a first article is required (see 3.2 and 6.1). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall be performed on each container. Inspection shall include the examination of 4.3 and the packaging inspection of 4.5.

4.3 Examination. Each container shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects (see table IV) preventing or lessening maximum efficiency shall constitute cause for rejection.

TABLE IV. Major defects.

Major defects	Requirement paragraph
Not of type or size specified	1.2
Limiting dimensions not as specified	3.4.1 and table I
Container opening sizes not as specified	3.4.2 and table II
Construction not as specified	3.5.1 thru 3.5.5
Reinforcements not as specified	3.6.2
Bottom discharge sections not as specified	3.6.3
Bottom discharge sump sections not as specified	3.6.4
Doors and latches not as specified	3.6.5 and 3.6.5.1
Hinges not as specified	3.6.6.1 thru 3.6.6.3
Boom rests not as specified	3.6.7
Lifting pins not as specified	3.6.8
Bail bars and assemblies not as specified	3.6.9
Casters not as specified	3.6.10
Workmanship not as specified	3.11

4.4 Tests. To verify compliance with the requirements of this specification, failure to pass any phase of the tests shall be cause for rejection. The first article refuse container shall be tested as follows:

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a. Place the employ refuse container on a level ground surface.

b. Load the container with a quantity of dry sand of the weight specified in table III for the applicable size container. The sand shall be deposited in the container so that the surface of the sand is approximately level. No container shall tip as a result of such loading.

c. A hoist-haul mechanism specified in MIL-T-11663, having a net payload capacity of not less than 9,000 lb, shall be utilized for testing of the containers.

d. The loaded refuse container shall be lifted by the hoisting mechanism to the proper height for engaging the automatic hook to the bail. The bail wall of the loaded container shall tilt back and bear against the hoist frame.

e. The container shall then be moved forward onto the container, rest in the hauling position and the booms lowered to rest on the boom rests. Each rest shall be in the proper position to equally support the boom and to permit hauling in this position.

f. The loaded container shall then be transported at a speed Of 20 miles per hour over an unimproved road for a distance of not less than one mile. The container shall balance and fit to the hoist-haul container rest and carriage.

g. The container shall then be examined for warpage, distortion, or malfunction, and if evidence of such defects is not found, the loaded container shall be lowered full speed from a height of two feet onto one 4 by 4 inch piece of wood placed horizontally on the ground so that the wood is in a diagonal position across the corners of the bottom of the container. The container shall then be lifted and examined for permanent distortion, warpage, bending, or other damage to any part of the container.

h. The bail of the container shall then be engaged by the automatic dumping hook, and the contents of the container emptied by use of the hoist-haul mechanism. Verify that the bail bar is readily engaged by the dumping hook, and that the doors and dumping features function properly and as specified. The tilt angle during the dumping operation shall be such that all material is dumped from container. Examine all parts of container for failure, warpage, or distortion.

i. For containers with sump, the sump shall be filled with water to the depth specified in 3.6.4. The container shall then be hoisted and placed in the haul position. A sump which leaks when in the level position or in the tilted hauling position, shall be cause for rejection of the container.

j. When casters are furnished, the loaded container shall be raised to a height of two feet and then lowered full speed onto a flat concrete surface for not less than 10 cycles. No breakage, distortion, cracking, or other malfunction of the caster assemblies and containers shall result. When the container is placed in the haul position, the caster shall not touch the fenders and other parts of the hoist-haul mechanism.

k. The self-balancing spring-hinged top doors shall be tested for conforance to 3.6.6.2.

l. Upon completion of the test specified in subparagraphs (a) through (h) above, load the container with trash common to commercial and residential pick up, and repeat operations (d) through (h). Inability to dislodge and empty all trash from inside the container without manual assistance shall be sufficient cause for rejection of the container.

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4.5 Packaging inspection. The packing and marking of the container shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Packing. Packing shall be level A or commercial as specified (see 6.1).

5.1.1 Level A. The refuse containers shall be shipped uncrated. All doors shall be closed and the spring-loaded latches secured with wire or other convenient peans to prevent accidental unlatching. Bottom discharge sections shall be secured with flat steel strapping conforming to ASTM D 3953, or by any other suitable means which will prevent opening of bottom sections during shipment. When strapping is used, fiberboard pads shall be placed under the strapping to prevent damage to the container.

5.1.2 Commercial. The equipment shall be prepared for shipment in accordance with ASTM D 3951.

5.2 Marking. Marking shall be in accordance with ASTM D 3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition retirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Make and model number of applicable hoist-haul mechanisms when container is procured for use with an existing refuse collection system (see 1.1 and 3.4).
c. Type and size of refuse container required (see 1.2).
d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
e. When first article is required for inspection and approval (see 3.1, 4.2.1, and 6.3).
f. When steel materials other than as specified are required (see 3.3.1)
g. When containers require sump (see 3.5.1 through 3.5.3).
h. When type II containers require top doors (see 3.5.2).
i. When type IV containers require top doors, and whether one or two tap doors are to be furnished (see 3.5.4).
j. When sliding end doors are required (see 3.6.5.1).
k. When type IV and V containers require bails suitable for use with old style single hook hoisting mechanism (see 3.6.9).
l. When casters are required (see 3.6.10).
m. Color of finish coat required (see 3.8).
n. Level of packing required (see 3.1).

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6.2 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.3 Refuse container types and variations. Figure 1 shows refuse containers which have been found acceptable; however, the figure is included for illustration only and not intended to preclude the furnishing of other designs which conform to this specification.

6.4 Subject term (key word) listing.

Rubbish receptacles
Trash cans
Waste receptacles

6.5 Part or identifying number (PIN). The PINs to be used for items acquired to this document are created as follows:

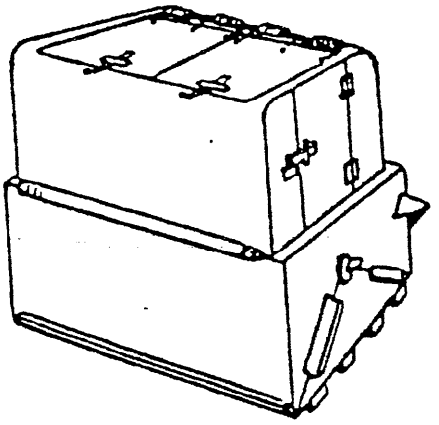
M	22827	XXXX
Military Spec	Spec No.	Code No.

The type and size of the containers are identified by a 4-digit code (see table V).

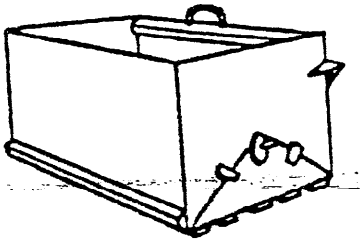
TABLE V. Type and size to code number.

Type and size	Code number	Type and size	Code number
Type I		Type IV	
Size 4	1040	Size 2-1/2	4025
Size 6	1060	Size 3	4030
Size 8	1080	Size 4	4040
Size 10	1100	Size 6	4060
Size 12	1120	Size 7-1/2	4075
Type II		Type V	
Size 4	2040	Size 4	5040
Size 6	2060	Size 6	5060
Size 8	2080	Size 8	5080
Type III			
Size 6	3060		
Size 8	3080		
Size 10	3100		

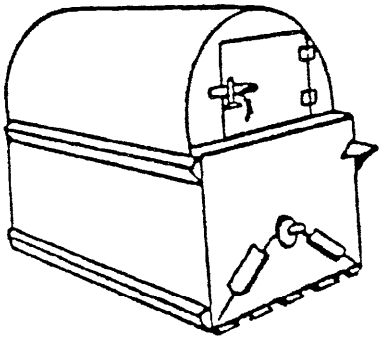
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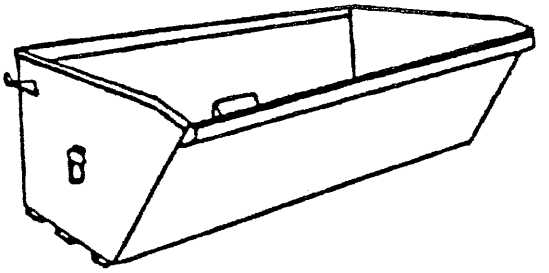
TYPE I
BOTTOM DISCHARGE
RECTANGULAR TOP



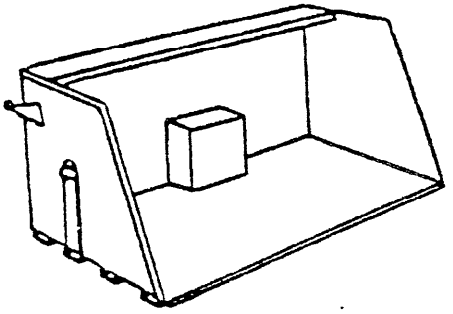
TYPE II
STANDARD BOTTOM DISCHARGE



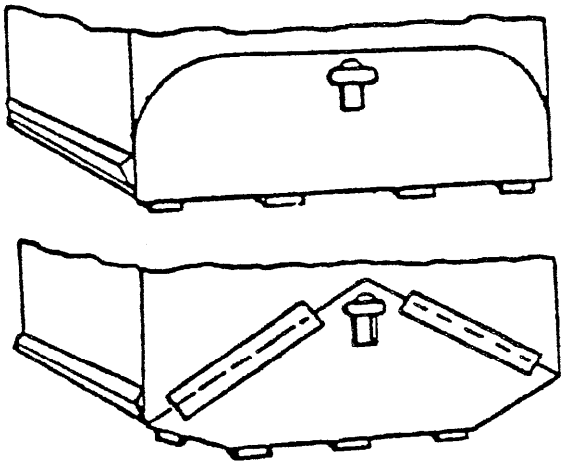
TYPE III
BOTTOM DISCHARGE,
ROUNDED TOP



TYPE IV
TOP DISCHARGE (TILT)



TYPE V
SKIP, FLAT BOTTOM



VARIATIONS OF BOTTOM
DISCHARGE AND SUMP
BOTTOM

FIGURE I. Refuse container types and variations

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-22827E	Refuse Container, Hoisting Units

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER (Specify):
b. ADDRESS (Street, City, State, ZIP Code)	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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